

5.17 Classwork: Logarithms Practice (no calculator)

1. [Maximum mark: 12]

Evaluate each of the following.

(a) $\log_4 \frac{1}{64}$

(b) $\log_{25} 5$

(c) $\log_2 \sqrt{32}$

(d) $\log_5 125$

(e) $\log_{16} \frac{1}{4}$

(f) $\log_3 \sqrt{81}$

[12]

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2. [Maximum mark: 12]

(a) Let $c = \log_3 m$, where $m > 0$. Write each of the following expressions in terms of c .

(i) $\log_3 m^2$

(ii) $\log_3(27m)$

(iii) $\log_9 m$ [6]

(b) Let $a = \log_2 m$, where $m > 0$. Write each of the following expressions in terms of a .

(i) $\log_2 m^3$

(ii) $\log_2(8m)$

(iii) $\log_4 m$ [6]

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3. [Maximum mark: 12]

Use the laws of logarithms.

(a) Express each in terms of $\log x$ and $\log y$.

(i) $\log(100x)$

(ii) $\log\left(\frac{x^2}{y}\right)$

(iii) $\log \sqrt{xy^3}$ [6]

(b) Write each expression as a single logarithm.

(i) $\log_2 x + \log_2 y$

(ii) $2 \log_7 a - \log_7 b$

(iii) $\frac{1}{2} \ln p + \ln q$ [6]

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4. [Maximum mark: 12]

Solve each equation. Give exact answers.

(a) $\log_2(x - 3) = \log_4(x + 9)$

(b) $\log_3 x = 2$

(c) $\log_5(2x) = 1$

(d) $\log_2(x+1) + \log_2(x-1) = 3$

[12]